

Rutuja Patel

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SUMMARY

Business Analytics graduate with hands-on experience in data engineering, data modeling, machine learning, and advanced analytics across healthcare, retail, manufacturing, and operations settings. Skilled in building ETL/ELT pipelines, data products, performance dashboards, and cloud-based analytical solutions (AWS) using Python, SQL, and industry-leading visualization tools to support data-driven decision-making. Experienced collaborating with cross-functional stakeholders and subject matter experts, including IT and business teams, to translate data into actionable insights that improve data quality, efficiency, and performance.

EDUCATION

W. P. Carey School of Business at Arizona State University Aug 2024 – Dec 2025
Master of Science in Business Analytics Tempe, AZ

- Relevant Coursework: AI and Data Analytics, Descriptive and Predictive Analytics, Analytics of Unstructured Data, Machine Learning in Business, Advanced Marketing Analytics, Enterprise Data Analytics, Operations Planning/Execution, Operations & Supply Chain Management, Quantitative Risk Management, Strategic Procurement, Logistics in the Supply Chain, Business Process Analytics

Gujarat Technological University July 2016 – July 2020
Bachelor of Engineering in Electronics and Communication Ahmedabad, GJ

PROFESSIONAL EXPERIENCE

Applied Analyst (Capstone + Internship) Aug 2025 – Present
DST Investment Advisors Scottsdale, AZ

- Built data mining and web-scraping pipelines (Playwright, Python) to extract and structure attorney lead data from multiple government sources into clean, analysis-ready data products, applying data governance, quality-control, and standardization best practices.
- Automated the end-to-end data-oriented programming workflow, significantly reducing manual research time and ensuring consistent, on-time data delivery for stakeholder reporting.

ERP Data Assistant Jan 2024 – May 2024
Hexaplast Industries (Electronics Manufacturing) Ahmedabad, GJ

- Compiled and structured data from multiple sources — office documents, sales contracts, spreadsheets, and real-time order data — supporting ERP implementation and organizational reporting standards at an electronics manufacturing facility.
- Queried and managed structured data using SQL, alongside Python and Excel, in an ETL workflow to clean, transform, and consolidate multi-source data (documents, sales contracts, real-time order data) supporting ERP implementation and cross-departmental reporting.

Business Operations Trainee May 2023 – Dec 2023
Hexaplast Industries (Electronics Manufacturing) Ahmedabad, GJ

- Completed cross-departmental training across sales, inventory, machine building, and electronics to build end-to-end operational and manufacturing knowledge ahead of specializing in data analytics.

PROJECT EXPERIENCE

Hospital Readmission of Diabetic Patients — ASU Data Science Lab Aug 2025 – Dec 2025

- Applied machine learning and data modeling techniques to a 101,766-record diabetic patient dataset from 130 U.S. hospitals to predict 30-day readmission risk, building and evaluating Logistic Regression, Random Forest, and XGBoost models to help improve patient care capabilities (0.672 AUC-ROC).
- Validated model robustness via 5-fold stratified cross-validation and supported responsible data governance by auditing 50 data fields and excluding race/gender from model inputs to mitigate bias.
- Presented methodology and findings verbally to a cross-disciplinary audience of data scientists, healthcare professionals, students, and librarians; published full technical documentation on a public portfolio site for transparency and reproducibility.

Diabetes Readmission Risk Assistant (RAG Application) Jan 2026 – Feb 2026

- Extended the readmission model into a proof-of-concept, live healthcare decision-support application, designing a retrieval-augmented generation (natural language processing) pipeline over 46 clinical guideline documents (ADA, WHO, CDC, Mayo Clinic) chunked into 2,456 embeddings to drive evidence-based decision-making.

- Integrated a live XGBoost risk score with an LLM (Llama-3.1-8B) via Streamlit, an industry-leading visualization and app framework, to deliver personalized, citation-backed care recommendations — directly relevant to optimizing healthcare operations and performance.

Inventory Optimization Using Computer Vision

Mar 2025 – May 2025

- Partnered with ASU's Pitchforks dining hall to automate a 6–8 hour manual inventory count, fine-tuning a YOLOv8 object-detection model via transfer learning on a self-collected 428-image dataset (96.8% precision, 0.973 mAP@0.5).
- Diagnosed a multi-object detection failure mode and resolved it using synthetic data augmentation, limiting the mAP@0.5 drop to just 0.011 — demonstrating applied research into innovative data analysis and modeling techniques.

Fast Fashion vs. Slow Fashion (Yelp Review Analysis)

Jan 2025 – Mar 2025

- Compared customer sentiment across retailers by cleaning and joining 5 raw Yelp datasets (~10M records) into a 65,729-review, 3,100-business dataset spanning 11 U.S. metros, staging and processing the data on AWS (S3) to manage its scale — a data mining and cloud-based data compilation effort from multiple sources.
- Ran dual natural language processing sentiment pipelines (VADER + DistilBERT) and applied TF-IDF/LDA topic modeling across 6 complaint categories, building an analytical framework with key metrics to inform business strategy recommendations.

Reviewer Social Network Analysis

Feb 2026 – Apr 2026

- Investigated whether Yelp's "Elite" reviewer status reflects genuine social influence by constructing a social graph of 46,236 reviewers and 135,151 friendships, applying graph-based data modeling techniques.
- Found that Veteran Elite reviewers were ~44x more socially connected than single-review users, using machine learning-based k-means geospatial clustering across 11 U.S. metros to support data-driven insights.

Promotion Impact & Causal Analysis

Sep 2026

- Applied Welch's t-test, Mann-Whitney U, and a store-level paired test to 1M+ daily retail sales records (1,115 stores), confirming a statistically significant promotional sales lift (38.8%, $p < 0.001$).
- Designed a two-way fixed-effects difference-in-differences model (store and time controls, clustered standard errors) to isolate the causal promotional lift at 39.3% (95% CI: 38.5–40.0%), backed by a SQL star-schema warehouse.

SKILLS & ACTIVITIES

- **Programming & Data-Oriented Languages:** Python, SQL, HTML, CSS, JavaScript
- **Cloud & Data Engineering:** AWS (S3), ETL/ELT Pipelines, Data Pipelines, Data Modeling, Data Governance & Quality Standardization
- **Analytics & Visualization Software:** Excel, Power BI, Tableau, Performance Dashboards
- **Machine Learning & Statistics:** Statistical Analysis, Hypothesis Testing, Causal Inference (Difference-in-Differences), Fixed-Effects Regression, scikit-learn, XGBoost, Random Forest, Logistic Regression, Data Modeling
- **AI & Generative AI:** Retrieval-Augmented Generation (RAG), LLMs (Llama-3.1-8B), Sentence-Transformers, ChromaDB, Streamlit
- **Deep Learning & Computer Vision:** YOLOv8, OpenCV, FastSAM, DistilBERT, Transfer Learning
- **NLP, Data Mining & Automation:** Sentiment Analysis (VADER), Topic Modeling (TF-IDF/LDA), NetworkX (Graph Analysis), Geospatial Clustering, Playwright (Web Automation), Data Governance & Quality Best Practices